Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	04 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form
		Registration through Gmail
		Registration through LinkedIn
FR-2	User Authentication	Login using Email and Password
		OTP Verification
		Forgot Password
FR-3	Weather Data Collection	Collect rainfall data
		Collect temperature data
		Collect humidity data
		Collect wind speed data
FR-4	Water Level Monitoring	Monitor river water levels
		Record flow rate
		Store historical water level data
FR-5	Data Pre-processing	Handle missing values
		Normalize and clean data
		Feature extraction
FR-6	Machine Learning Model	Train flood prediction model
		Test model accuracy
		Predict flood probability
FR-7	Flood Risk Prediction	Classify risk as Low, Medium, or High
		Generate prediction report

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide a simple, intuitive, and user-friendly interface that allows users to access flood predictions, alerts, and reports with minimal training.
NFR-2	Security	The system should ensure secure user authentication, encrypt sensitive data, protect against unauthorized access, and maintain data privacy.
NFR-3	Reliability	The system should generate accurate flood predictions, maintain data integrity, and operate consistently with minimal failures.
NFR-4	Performance	The system should process weather and water-level data efficiently and generate flood prediction results and alerts within a few seconds of receiving new data.
NFR-5	Availability	The system should be available 24×7 with high uptime to ensure continuous monitoring and timely flood alerts during emergencies.
NFR-6	Scalability	The system should support increasing numbers of users, monitoring stations, and data sources without significant degradation in performance.